Cantera vs NN | Test IC 53 | NRMSE (util.py)

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
2 P-only DeepONet10	**10k**	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	**long**	6.02e-03	9.36e-03	0.124	0.107	0.0969	0.04	3.93e-03	0.253	0.156	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	**long**	5.73e-03	9.72e-03	0.138	0.182	0.0662	0.0114	2.73e-03	0.158	0.109	0	Complete
4 T/P DeepONet10	**10k**	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	**Stop 0.486 ms**
5 ANN10 + T/P DeepONet10	**10k**	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0321	0.0129	0.439	0.247	0.0231	0.558	0.0178	0.139	0.34	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.653	0.451	10.4	0.707	0.052	6.82	0.212	59.6	8.41e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.376	0.201	5.52	1.2	0.0595	5.41	0.167	0.159	0.353	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.